Nguyen Song Thien Long

Publication name: Long S. T. Nguyen

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Research Interests

My research interests lie in **Natural Language Processing (NLP)**, with a primary focus on Information Retrieval (IR), Question Answering (QA), and Large Language Models (LLMs). My work centers on Retrieval-Augmented Generation (RAG), hybrid lexical–semantic retrieval, and knowledge graph-grounded, multi-hop reasoning in both agentic and standalone LLM systems. I am especially interested in making these methods efficient and trustworthy, including in low-resource and multilingual settings.

Employment

Apr 2026 – Present
AI Researcher. Saigon AI Hub, VNG Group.
Serving as *Principal Investigator* for a granted research project on structured reasoning and training strategies for LLMs, funded by VNG Group and Vietnam National University Ho Chi Minh City (VNU-HCM).

Oct 2025 – Present
AI Researcher. Center for AI Research (CAIR), VinUniversity.
Investigating the training of lexical–dense and hybrid retrieval models and their integration into LLM-based QA systems, mentored by Prof. Luu Anh Tuan (NTU, Singapore) and Dr. Nguyen Tran Cong Duy (CAIR).

Dec 2023 – Present
Research Assistant. URA Research Group, Faculty of Computer Science and Engineering, HCMUT.
Conducting research on IR, QA, and LLM-based systems under the supervision of Prof. Quan Thanh Tho, and serving as *Team Leader*, defining research directions and mentoring junior researchers.

Education

2023 – 2027
B.Eng. in Computer Science (High-Quality Program), Ho Chi Minh City University of Technology (HCMUT), VNU-HCM, Vietnam. CGPA: 3.6/4.0.

Selected Publications (full list at https://longstnguyen.com/publications/)

2027
- Thi Ty Nguyen, **Long S. T. Nguyen**, Tan Sang Nguyen, Nhu Tran, Khang Nguyen, Dong Doan, Vy Le, Ngoc Tran, Phat T. Tran-Truong, and Tho T. Quan. *Fluent but Foreign: Benchmarking Cultural Understanding of Multilingual Large Language Models in Vietnamese.* ACM International Conference on Web Search and Data Mining (**WSDM**). (CORE A)
- **Long S. T. Nguyen**, Cong-Duy T. Nguyen, Anh Tuan Luu, Vu Duong, and Tho T. Quan. *Sentence Matching Is Not Enough: Relational Geometry Transfer for Post-Hoc Language Expansion of Retrievers.* ACM International Conference on Web Search and Data Mining (**WSDM**). (CORE A)
- **Long S. T. Nguyen**. *English-Only Adaptation Splits Monolingual and Cross-Lingual Retrieval: Task-Axis Dissociation and Retention Trade-offs.* ACM International Conference on Web Search and Data Mining (**WSDM**). (CORE A)

2026
- **Long S. T. Nguyen**, Dat T. Huynh, Quynh T. N. Vo, Dang Le, Cong-Duy T. Nguyen, Anh Tuan Luu, Anh Nguyen-Duc, and Tho T. Quan. *Fast and Focused: Interactive Repository-Level Fill-in-the-Middle Code Completion with Hybrid Structural Retrieval.* Conference on Empirical Methods in Natural Language Processing (**EMNLP**). (CORE A*)
- Thi Ty Nguyen, **Long S. T. Nguyen**, Phat T. Tran-Truong, Tan Sang Nguyen, and Tho T. Quan. *Bahnar–Vietnamese Cross-Lingual Sentence Retrieval: A Low-Resource Case Study with Lexical, Neural, and Hybrid Methods.* Asia-Pacific Chapter of the Association for Computational Linguistics & International Joint Conference on Natural Language Processing (**AACL-IJCNLP**).
- **Long Nguyen**, Duc Nguyen, Quan Bui, Dung Phan, Dung Le, Khanh Nguyen, Quynh Vo, Khang Vo, Nam Duong, Anh Dinh, Tri Trinh, Chi Phan, An Nguyen, Thai Nguyen, Dang Le, Vinh Dang, and Tho Quan. *Catching the First Light of Tomorrow: A Hackathon-Based Framework for Introducing High School Students to AI Agents.* AAAI Symposium on Educational Advances in Artificial Intelligence (**EAAI**). (CORE A*)
- **Long S. T. Nguyen** and Tho T. Quan. *Which Works Best for Vietnamese? A Practical Study of Information Retrieval Methods across Domains.* Findings of the Association for Computational Linguistics: **EACL**. (CORE A)
- Hung C. Luu, **Long S. T. Nguyen**, Trung M. Pham, Hieu M. Pham, and Tho T. Quan. *HiGraAgent: Dual-Agent Adaptive Reasoning over Hierarchical Knowledge Graph for Open-Domain Multi-hop Question Answering.* Findings of the Association for Computational Linguistics: **EACL**. (CORE A)
- Quoc Phu Dang*, Phat T. Tran-Truong*, Duc-Ly Vu, **Long S. T. Nguyen**, Quynh T. N. Vo, and Tho T. Quan. *Enhancing Large Language Model Performance for Automatic Zero-Shot Multiple-Choice Question Answering via Single-Token Logit Prompting.* Computers and Education: Artificial Intelligence. (SCIE, Q1, #1 IN AI)
- Mao Nguyen, Thien Pham, **Long S. T. Nguyen**, Dang Le, Trung M. Pham, Hieu M. Pham, Duc Q. Nguyen, Xinh Le, Anh H. Huynh, Triet T. M. Le, and Tho T. Quan. *Conceptual Relationship Between Classical Machine Learning and Graph Neural Networks in Out-of-Distribution Detection: A Comprehensive Survey.* **IEEE Access**. (SCIE, Q1)

2025
- **Long S. T. Nguyen**, Quynh T. N. Vo, Hung C. Luu, and Tho T. Quan. *When in Doubt, Ask First: A Unified Retrieval Agent-Based System for Ambiguous and Unanswerable Question Answering.* Findings of the Association for Computational Linguistics: **IJCNLP-AACL**.
- Duc Nguyen, Dong Le, **Long Nguyen**, Quyen Vu, Tran Le, Dung Nguyen, Nga Huynh, Huong Nguyen, Phat Tran, Dang Le, Sang Truong, Sanmi Koyejo, Cuong Le, and Tho Quan. *Riding on The Back of A Whale: A Hackathon Framework for Introducing High School Students to Large Language Models.* Conference on Artificial Intelligence in Education (**AIED**). (CORE A)

Selected Publications (full list at https://longstnguyen.com/publications/) (continued)

- **Long S. T. Nguyen**, Tran T. B. Le, Huong P. N. Nguyen, Quynh T. N. Vo, Phong H. N. Nguyen, and Tho T. Quan. *Serving the Underserved: Leveraging BARTBahnar Language Model for Bahnaric-Vietnamese Translation.* Workshop on Language Models for Underserved Communities (**LM4UC**), at **NAACL**. (WORKSHOP, CORE A)

Academic Service

Prog. Cmte.	AAMAS 2027 (CORE A); AAAI/EAAI 2027 (CORE A*); ACM WSDM 2027 (CORE A).
Conf. Rev.	WWW 2027 (CORE A*); ACM MM 2026 (CORE A*).
Journal Rev.	IEEE Transactions on Neural Networks and Learning Systems (Q1); Expert Systems with Applications (Q1); Engineering Applications of Artificial Intelligence (Q1); IEEE Access (Q1); Behaviour & Information Technology (Q1); Journal of Natural Language Processing (Q1).
Olympiad	Technical and training support for VOAI 2025 and the preparation of Vietnam's IOAI 2025 team.

References

Supervisor	**Prof. Quan Thanh Tho.** Dean, Faculty of Computer Science and Engineering, HCMUT. qttho@hcmut.edu.vn
Mentor	**Prof. Luu Anh Tuan.** Nanyang Technological University (NTU), Singapore. anhtuan.luu@ntu.edu.sg